Assignment No.: 02

Name: Nikita Lalaso Narute **Roll No.: COB422**

Batch: B2 **Div: B**

Class: BE COMPUTER

Title:

Write a program to implement Huffman Encoding using a greedy strategy.

Program:

```
#include <iostream>
#include <queue>
#include <vector>
#include <string>
#include <unordered_map>

using namespace std;

struct Node {
    char ch;
    int freq;
    Node *left, *right;

    Node(char ch, int freq) {
        left = right = nullptr;
        this->ch = ch;
        this->freq = freq;
    }
};

struct compare {
    bool operator()(Node* l, Node* r) {
        return l->freq > r->freq;
    }
};

void printCodes(struct Node* root, string str) {
    if (!root)
        return;

    if (root->ch != '$')
        cout << root->ch << ": " << str << "\n";

    printCodes(root->left, str + "0");
    printCodes(root->right, str + "1");
}

void HuffmanCodes(char data[], int freq[], int size) {
    struct Node *left, *right, *top;
    priority_queue<Node*, vector<Node*>, compare> minHeap;

    for (int i = 0; i < size; ++i)
        minHeap.push(new Node(data[i], freq[i]));

    while (minHeap.size() != 1) {
        left = minHeap.top();
        minHeap.pop();

        right = minHeap.top();
        minHeap.pop();

        top = new Node('$', left->freq + right->freq);
        top->left = left;
        top->right = right;
        minHeap.push(top);
    }

    printCodes(minHeap.top(), "");
}

int main() {
    char arr[] = {'a', 'b', 'c', 'd', 'e', 'f'};
    int freq[] = {5, 9, 12, 13, 16, 45};

    int size = sizeof(arr) / sizeof(arr[0]);
    HuffmanCodes(arr, freq, size);

    return 0;
}
```

Output:

```
(base) sspm@sspm:~$ g++ daa2.cpp
(base) sspm@sspm:~$ ./a.out
f: 0
c: 100
d: 101
```

a: 1100
b: 1101
e: 111